

Mechanical Engineering: Stress Tensors, Mohrs Circle & Power Cycles

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1. Stress Transformation & Mohrs Circle

State of 2D plane stress is governed by normal stresses (σ_x , σ_y) and shear stress (τ_{xy}).
Mohrs circle graphically represents principal stresses (σ_1 , σ_2) and max shear (τ_{max}).
Principal planes experience zero shear stress and maximum tensile/compressive normal loads.

2. Applied Thermodynamics & Air Standard Cycles

First Law establishes conservation of energy ($dQ = dU + dW$); Second Law governs entropy growth.
Otto Cycle (constant volume heat addition) models spark-ignition internal combustion engines.
Diesel Cycle (constant pressure heat addition) achieves higher compression ratios.

3. Mechanism Kinematics & Linkages

Grashofs Law determines whether a four-bar linkage has at least one continuously rotating crank.
Degree of freedom (mobility) is computed using Kutzbach criterion.